

■ VOLATILITY CALCULATION GUIDE

TECU Credit Union | Contact Center Analytics

1. What is Volatility?

Volatility = A measure of how much a metric **varies or fluctuates** over time.

It answers the question: '**How stable/consistent is this metric?**'

Simple Definition: Volatility shows if your data is predictable (stable) or bouncy (unpredictable).

2. The Math Behind Volatility

Standard Deviation Formula:

$\sigma = \sqrt{[\Sigma(x - \mu)^2 / N]}$	Where:
	σ = Standard Deviation (volatility measure)
	x = Each data point
	μ (mu) = Average/Mean of all data points
	N = Number of data points
	Σ = Sum of all values

3. Worked Example: Sentiment Volatility

Scenario: 6 Weeks of Sentiment Data

Week 1	72%
Week 2	74%
Week 3	75%
Week 4	76%
Week 5	77%
Week 6	78%

Step 1: Calculate the Average (Mean)

Average = $(72 + 74 + 75 + 76 + 77 + 78) \div 6$
Average = $452 \div 6$
Average = **75.33%**

Step 2: Find Difference from Average for Each Week

Week	Value	Difference from Average
Week 1	72%	$72 - 75.33 = -3.33$
Week 2	74%	$74 - 75.33 = -1.33$
Week 3	75%	$75 - 75.33 = -0.33$
Week 4	76%	$76 - 75.33 = +0.67$
Week 5	77%	$77 - 75.33 = +1.67$
Week 6	78%	$78 - 75.33 = +2.67$

Step 3: Square Each Difference

Difference	Squared
-3.33	$(-3.33)^2 = 11.09$
-1.33	$(-1.33)^2 = 1.77$
-0.33	$(-0.33)^2 = 0.11$
+0.67	$(+0.67)^2 = 0.45$
+1.67	$(+1.67)^2 = 2.79$
+2.67	$(+2.67)^2 = 7.13$
SUM	23.34

Step 4: Calculate Standard Deviation

Standard Deviation = $\sqrt{(23.34 \div 6)}$
Standard Deviation = $\sqrt{3.89}$
Standard Deviation = **1.97%**

✓ **RESULT: Volatility = 1.97% (VERY LOW)**

4. Volatility Scale: How to Interpret

Standard Deviation	Volatility Level	What It Means
0% - 2%	■ VERY LOW	Extremely stable
2% - 5%	■ LOW	Stable, consistent
5% - 10%	■ MEDIUM	Some variation
10% - 15%	■ HIGH	Significant swings
15%+	■ VERY HIGH	Highly unpredictable

5. Quick Reference: How to Calculate

Step	What to Do	Example
1	Collect data points	Sentiment: 72%, 74%, 75%, 76%, 77%, 78%
2	Calculate average	$(72+74+75+76+77+78) \div 6 = 75.33\%$
3	Find deviation from average	$72-75.33 = -3.33$, $74-75.33 = -1.33$, etc.
4	Square each deviation	$(-3.33)^2=11.09$, $(-1.33)^2=1.77$, etc.
5	Find average of squared deviations	$\text{Sum} \div \text{count} = 23.34 \div 6 = 3.89$
6	Take square root	$\sqrt{3.89} = 1.97\%$
7	Interpret	$1.97\% = \text{LOW} \checkmark$

6. Key Takeaways for Contact Centers

Scenario	Volatility	Status	Action
Stable sentiment (72→78%)	1.97%	✓ LOW	Keep doing what you're doing
Unstable sentiment (95→52%)	17.25%	✗ HIGH	Investigate the cause
Consistent quality scores	2-3%	✓ LOW	Predictable & manageable
Variable quality scores	12%+	✗ HIGH	Training & coaching needed